

→JS-9

DOM (Document Object Model)

The DOM **represents a document** with a logical tree.

It allows us to manipulate/change webpage content (HTML elements).



parent ←
↓
child ←

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→window object is inbuilt it will automatically generate in console

→Document

```
> window
< Window {0: Window, window: Window, self: Window, document: document, name: '', location: Location, ...}

> document
< #document
  <!DOCTYPE html>
  <html dir="ltr" lang="en" class lazy-loaded="true">
    <head> ... </head>
    <body style="background-color: rgb(53, 54, 58);">
      ... </body>
  </html>
```

→console.log(document) we will get same above result

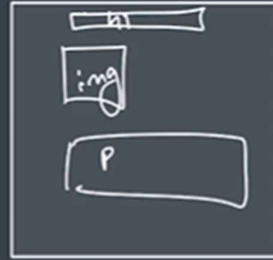
```
> console.dir(document);
VM361:1
#document 1
  location: Location {ancestorOrigins: DOMStringList, URL: "chrome://new-tab-page/"
  activeElement: body
  adoptedStyleSheets: Proxy(Array) {}
  alinkColor: ""
  all: HTMLAllCollection(139) [html, head, meta, title, ...]
  anchors: HTMLCollection []
  applets: HTMLCollection []
  baseURI: "chrome://new-tab-page/"
  bgColor: ""
  body: body
    characterSet: "UTF-8"
    charset: "UTF-8"
    childElementCount: 1
  childNodes: NodeList(2) [<!DOCTYPE html>, html]
  children: HTMLCollection [html]
  compatMode: "CSS1Compat"
  contentType: "text/html"
  cookie: ""
  currentScript: null
```

Download Starter Code

1) Select (img)

2) changes/manipulation

JS



→ selection and changes/manipulation will be done by js

Selecting Elements

getElementById

Returns the Element as an object or null (if not found)

The screenshot shows a web browser with a page titled "Subbu" featuring a Spider-Man image and an "About" section. The "About" section describes Spider-Man as a superhero appearing in American comic books published by Marvel Comics, created by writer-editor Stan Lee and artist Steve Ditko. It mentions his first appearance in the anthology comic book *Amazing Fantasy* #15 (August 1962) in the *Silver Age of Comic Books*. The "Publication Info" section lists the Publisher as Marvel Comics, the First Appearance as *Amazing Fantasy* #15, and the Created by as Stan Lee.

The Chrome DevTools Console is open, showing the following code and output:

```
> console.dir(document.all);
  ▶ HTMLAllCollection(39) VM116:1
< undefined
> console.dir(document.all[8]);
  ▶ h1 VM230:1
< undefined
> console.dir(document.all[8].innerText);
  Spider Man VM344:1
< undefined
> document.all[8].innerText="Subbu";
< 'Subbu'
>
```

The console also shows a message: "Failed to load resource: the server responded with a status of 404 (Not Found)" for the file `index.html:92` and `5500/favicon.ico:1`.

Dimensions: Re 275 x 653 10 DPF

Spider Man



About

Spider-Man is a superhero appearing in American comic books published by [Marvel Comics](#). Created by writer-editor **Stan Lee** and artist **Steve Ditko**, he first appeared in the anthology comic book [Amazing Fantasy](#) #15 (August 1962) in the [Silver Age of Comic Books](#). He has been featured in comic books, television shows, films, video games, novels, and plays

Publication Info

- [Publisher](#)
- [First Appearance](#)
- Created by
 - [Stan Lee](#)

Console

top

Filter

Default levels

No Issues

☒ Network messages

☐ Preserve log

☐ Selected context only

☒ Group similar messages

☒ CORS errors in console

☐ Log XMLHttpRequests

☒ Eager evaluation

☒ Autocomplete from history

☒ Treat code evaluation as user action

> document.getElementById("mainImg");

<

> let imObj=document.getElementById("mainImg");

< undefined

> imObj

<

> console.log(IMO

Uncaught SyntaxError: missing) after argument list VM714:1

> console.log(imObj);

< VM739:1

< undefined

> console.dir(imObj);

VM750:1

img#mainImg

accessKey: ""

activeViewTransition: null

align: ""

alt: ""

ariaActiveDescendantElement: null

ariaAtomic: null

ariaAutoComplete: null

ariaBrailleLabel: null

ariaBrailleRoleDescription: null

ariaBusy: null

ariaChecked: null

ariaColCount: null

ariaColIndex: null

ariaColIndexText: null

ariaColSpan: null

ariaControlsElements: null

ariaCurrent: null

ariaDescribedByElements: null

Console

Issues

Spider Man



About

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- [First Appearance](#)
- Created by
 - [Stan Lee](#)

- ☒ Network messages
- ☐ Preserve log
- ☐ Selected context only
- ☒ Group similar messages
- ☒ CORS errors in console
- ☐ Log XMLHttpRequests
- ☒ Eager evaluation
- ☒ Autocomplete from history
- ☒ Treat code evaluation as user action

```
Uncaught SyntaxError: missing ) after argument list VM141:1

> console.log(imObj);
     VM739:1
< undefined
> console.dir(imObj);
  ▶ img#mainImg VM750:1
< undefined
> imObj.src
< 'http://127.0.0.1:5500/js-9/assets/spiderman_img.png'
> imObj.tagName;
< 'IMG'
> console.dir(document.getElementById("description"));
Uncaught SyntaxError: missing ) after argument list VM1055:1
> console.dir(document.getElementById("description"));
VM1060:1

▼ #description
  accessKey: ""
  activeViewTransition: null
  align: ""
  ariaActiveDescendantElement: null
  ariaAtomic: null
  ariaAutoComplete: null
  ariaBrailleLabel: null
  ariaBrailleRoleDescription: null
  ariaBusy: null
  ariaChecked: null
  ariaColCount: null
  ariaColIndex: null
  ariaColIndexText: null
  ariaColSpan: null
  ariaControlsElements: null
  ariaCurrent: null
```

File | /Users/shradhakhapra/WebClassroom/JS/JavaScript%20Classroom/index.html

About

Spider-Man is a superhero appearing in American comic books published by [Marvel Comics](#). Created by writer-editor **Stan Lee** and artist **Steve Ditko**, he first appeared in the anthology comic book [Amazing Fantasy](#) #15 (August 1962) in the [Silver Age of Comic Books](#). He has been featured in comic books, television shows, films, video games, novels, and plays

Elements Console Sources Network >>
top Filter Default levels No Issues
> document.getElementById("abc");
< null

Selecting Elements

getElementByClassName

Returns the Elements as an HTML Collection or empty collection (if not found)


```
document.getElementsByClassName("oldImg");
HTMLCollection(3) [img.oldImg, img.oldImg, img.oldImg]
  0: img.oldImg
  1: img.oldImg
  2: img.oldImg
  length: 3
  [[Prototype]]: HTMLCollection
```

```
document.getElementsByClassName("oldImg")[0];

```

```
app.js > ...
let smallImages = document.getElementsByClassName("oldImg");

for (let i = 0; i < smallImages.length; i++) {
  console.dir(smallImages[i]);
}
```

```
img.oldImg app.js:4
img.oldImg app.js:4
img.oldImg app.js:4
```

```
1 let smallImages = document.getElementsByClassName("oldImg");
2
3 for (let i = 0; i < smallImages.length; i++) {
4   console.dir(smallImages[i].src);
5 }
6
```

```
file:///Users/shradhakhapra/WebClassroom/JS/JavaScript%20Cl app.js:4
assroom/assets/creation_1.png
file:///Users/shradhakhapra/WebClassroom/JS/JavaScript%20Cl app.js:4
assroom/assets/creation_2.jpeg
file:///Users/shradhakhapra/WebClassroom/JS/JavaScript%20Cl app.js:4
assroom/assets/creation_3.jpeg
```

```
app.js > ...
let smallImages = document.getElementsByClassName("oldImg");

for (let i = 0; i < smallImages.length; i++) {
  smallImages[i].src = "assets/spiderman_img.png";
  console.log(`value of image no. ${i} is changed.`);
}
```


Spider-Man

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File /Users/shradhakapra/WebClassroom/JS/JavaScript%20Classroom/index.html

Elements Console Sources Network

top Filter Default levels No Issues

```
> document.getElementsByTagName("p");
< ▶ HTMLCollection(2) [p, p#description, description: p#description]
> document.getElementsByTagName("p")[1].innerText = "abc";
< 'abc'
```

ABC

Publication Info

- Publisher
- First Appearance
- Created by
 - Stan Lee
 - Steve Ditko

Creation & Development

abc

```
> document.getElementsByTagName("span");
< ▶ HTMLCollection []
```

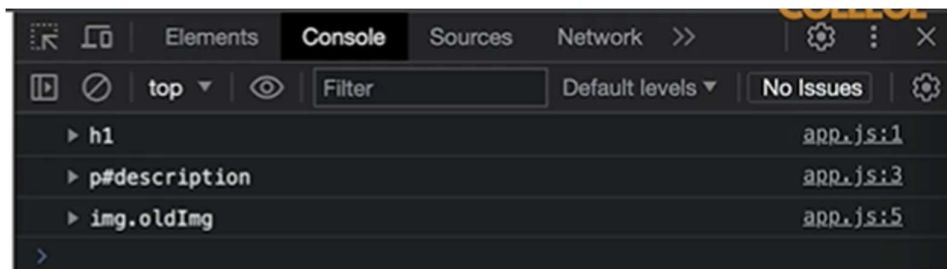
Query Selectors

Allows us to use any CSS selector

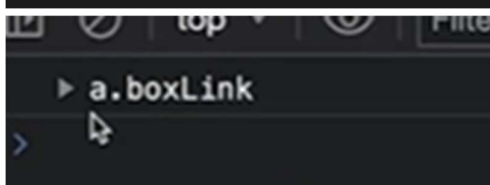
className: nth-of-type(2n+1)

```
document.querySelector('p'); //Selects first p element
document.querySelector('#myId'); //Selects first element with id = myId
document.querySelector('.myClass'); //Selects first element with class = myClass
document.querySelectorAll("p"); //Selects all p elements
```

```
1 console.dir(document.querySelector("h1"));
2
3 console.dir(document.querySelector("#description"));
4
5 console.dir(document.querySelector(".oldImg"));
6
```



```
console.dir(document.querySelector("div a"));|
```



```
console.dir(document.querySelectorAll("div a"));
```



Using Properties & Methods

innerText

Shows the visible text contained in a node

textContent

Shows all the full text

innerHTML

Shows the full markup

Manipulating Attributes

`obj.getAttribute( attr )`

`obj.setAttribute( attr, val )`

Spider Man

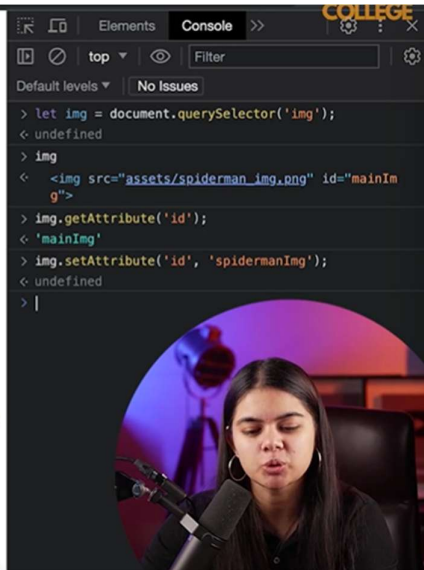


About

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Publication Info

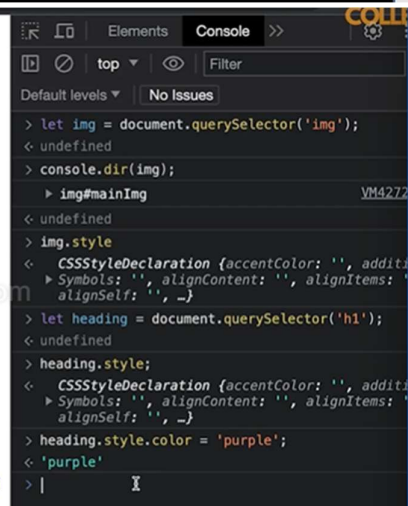
- Publisher
- First Appearance
- Created by
 - Stan Lee



Spider Man



books published by . Created by writer-editor Stan Lee and artist Steve Ditko, he first appeared in the anthology comic book *Amazing Fantasy* #15 (August 1962) in the *Silver Age of Comic Books*. He has been featured in comic books, television shows, films, video games, novels, and plays



```
1 let links = document.querySelectorAll(".box a");
2
3 for (let i = 0; i < links.length; i++) {
4   links[i].style.color = "yellow";
5 }
6
```

[Comic Books](#). He has been featured in comic books.

Publication Info

- [Publisher](#)
- [First Appearance](#)
- Created by
 - [Stan Lee](#)
 - [Steve Ditko](#)

Manipulating Style

using `classList`

`obj.classList`

`classList.add()` to add new classes

`classList.remove()` to remove classes

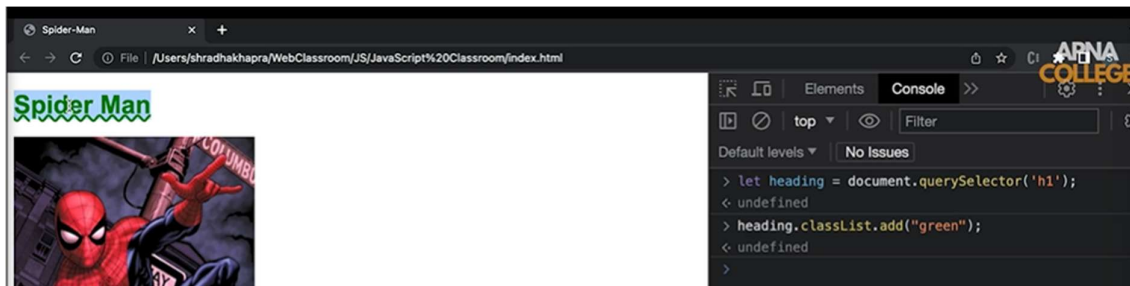
`classList.contains()` to check if class exists

`classList.toggle()` to toggle between add & remove

```
style.css > .green
body {
  font-family: Arial;
}

.green {
  color: green;
  text-decoration: wavy;
}
```

Produces a wavy line



```
> let heading = document.querySelector('h1');  
< undefined  
> heading.classList.add('green');  
< undefined  
> heading.classList.add('underline');  
< undefined  
> heading.classList.remove('green');  
< undefined  
> heading.classList.remove('green');  
< undefined  
> heading.setAttribute('class', 'green');  
< undefined  
> heading.classList  
< ▶ DOMTokenList ['green', value: 'green']  
> heading.classList.contains("underline");  
< false  
> heading.classList.contains("green");  
< true  
>
```

```
> heading.classList  
< ▶ DOMTokenList ['green', value: 'green']  
> heading.toggle("green");  
✖ ▶ Uncaught TypeError: heading.toggle is VM6399:1  
not a function  
at <anonymous>:1:9  
> heading.classList.toggle("green");  
< false  
> heading.classList  
< ▶ DOMTokenList [value: '']  
> heading.classList.toggle("underline");  
< true  
> heading.classList.toggle("green");  
< true  
> heading.classList  
< ▶ DOMTokenList(2) ['underline', 'green', value:  
'underline green']  
> heading.classList.toggle("underline");  
< false  
> heading.classList  
< ▼ DOMTokenList ['green', value: 'green'] ⓘ  
  0: "green"  
  length: 1
```

Navigation

parentElement

children

previousElementSibling / nextElementSibling

```
Default levels: NO ISSUES
> let h4 = document.querySelector('h4');
< undefined
> h4.parentElement
< ><div class="box"> ... </div>
> h4.children
< >HTMLCollection []
> let box = document.querySelector('.box');
< undefined
> box.children;
< >HTMLCollection(2) [h4, ul]
> box.childElementCount;
< 2
> let ul = document.querySelector('ul');
< undefined
> ul.parentElement
< ><div class="box"> ... </div>
> ul.childElementCount;
< 3
> ul.children;
< >HTMLCollection(3) [li, li, li]
  > 0: li
  > 1: li
  > 2: li
  length: 3
  > [[Prototype]]: HTMLCollection
```

```
Default levels ▾ | No Issues

> ul
< ▶ <ul> ... </ul>

> ul.children
< ▶ HTMLCollection(3) [li, li, li]

> ul.children[0];
< ▶ <li> ... </li>

> ul.children[1];
< ▶ <li> ... </li>

> ul.children[2];
< ▶ <li> ... </li>

> ul.children[2].previousElementSibling;
< ▶ <li> ... </li>

> ul.children[1].previousElementSibling;
< ▶ <li> ... </li>

> ul.children[1].nextElementSibling;
< ▶ <li> ... </li>

> ul.children[1].previousElementSibling;|
< li
```

Spider Man



```
APNA COLLEGE
Elements Console >>
top ▾ Filter
Default levels ▾ | No Issues

> let img = document.querySelector('img');
< undefined

> img.previousElementSibling;
< <h1>Spider Man</h1>

> img.previousElementSibling.style;
< CSSStyleDeclaration {accentColor: '', additive
  ▶ Symbols: '', alignContent: '', alignItems: '',
    alignSelf: '', -}

> img.previousElementSibling.style.color = "green";
< 'green'

>
```


Adding Elements

`document.createElement('p')`

- `appendChild( element )`
- `append( element )`
- `prepend( element )`
- `insertAdjacent( where, element)`

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Publication Info

- Publisher
- First Appearance
- Created by
 - Stan Lee
 - Steve Ditko

Creation & Development

In 1962, with the success of the *Fantastic Four*, Marvel Comics editor and head writer Stan Lee was casting for a new superhero idea. He said the idea for Spider-Man arose from a surge in teenage demand for comic books and the desire to create a character with whom teens could identify. As with *Fantastic Four*, Lee saw Spider-Man as an opportunity to "get out of his system" what he felt was missing in comic books. In his autobiography, Lee cites the non-superhuman pulp magazine crime fighter the Spider as a great influence, and in a multitude of print and video interviews, Lee stated he was further inspired by seeing a spider climb up a wall—adding in his autobiography that he has told that story so often he has become unsure of whether or not this is true.



Hi, I am a new p

APNA COLLEGE

Default levels ▾ No Issues

```
> document.createElement('p');
< <p></p>
> let newP = document.createElement('p');
< undefined
> console.dir(newP);
  p VM8278:
< undefined
> newP.innerText = "Hi, I am a new p";
< 'Hi, I am a new p'
> console.dir(newP);
  p VM8376:
< undefined
> let body = document.querySelector('body');
< undefined
> body.appendChild(newP);
< <p>Hi, I am a
>
```

```
> document.createElement('p');
< <p></p>
> let newP = document.createElement('p');
< undefined
> console.dir(newP);
  ▶ p VM8278:1
< undefined
> newP.innerText = "Hi, I am a new p";
< 'Hi, I am a new p'
> console.dir(newP);
  ▶ p VM8376:1
< undefined
> let body = document.querySelector('body');
< undefined
> body.appendChild(newP);
< <p>Hi, I am a new p</p>
>
```

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 - Steve Ditko

Hi, I am a new p

Creation & Development

In 1962, with the success of the Fantastic Four, Marvel Comics editor and head writer Stan Lee was casting for a new superhero idea. The idea for Spider-Man arose from a surge in teenage demand for comic books and the desire to create a new, relatable superhero identity. As with Fantastic Four, Lee saw Spider-Man as an opportunity to "get out of the superhero mold" and create a character with a more human-like personality. In his autobiography, Lee cites the non-superhuman pulp magazine *Detective Comics* as an influence, and in a multitude of print and video interviews, Lee stated he was inspired by the character's "relatable" nature—adding in his autobiography that he has told that story so often he has



```
Elements Console
top Filter
Default levels No Issues
> newP
< <p>Hi, I am a new p</p>
> let box = document.querySelector('.box');
< undefined
> box.appendChild(newP);
< <p>Hi, I am a new p</p>
> let btn = document.createElement('button');
< undefined
> console.dir('btn');
  btn VM8844:1
< undefined
> console.dir(btn);
  ▶ button VM8852:1
< undefined
> btn.innerText = "click me!";
< 'click me!'
> box.appendChild(btn);
< <button>click me!</button>
>
```

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Hi, I am a new p this is new text

Creation & Development

```
Elements Console
top Filter
Default levels No Issues
> newP
< <p>Hi, I am a new p</p>
> newP.append("this is new text");
< undefined
> newP.append(btn);
< undefined
> newP.append(" do not click this button");
< undefined
>
```

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Hi, I am a new pthis is new text click me! do not click this button

Elements Console >>

top Filter

Default levels No Issues

```

> newP
< <p>Hi, I am a new p</p>
> newP.append("this is new text");
< undefined
> newP.append(btn);
< undefined
> newP.append(" do not click this button");
< undefined
>

```

← → ↻ 📁 📄 📄 | /Users/shradhakapra/WebClassroom/JS/JavaScript%20Classroom/index.html



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NEW BUTTON!!!

Elements Console >>

top Filter

Default levels No Issues

```

> btn
< <button>NEW BUTTON!!!</button>
> p
< <p>< </p>
> p.insertAdjacentElement('beforebegin', btn);
< <button>NEW BUTTON!!!</button>
> p.insertAdjacentElement('afterbegin', btn);
< <button>NEW BUTTON!!!</button>
> p.insertAdjacentElement('beforeend', btn);
< <button>NEW BUTTON!!!</button>
> p.insertAdjacentElement('afterend', btn);
< <button>NEW BUTTON!!!</button>
> |

```

Removing Elements

- `removeChild( element )`
- `remove( element )`

```
< <button>NEW BUTTON!!!</button>
> let body = document.querySelector('body');
< undefined
> body.removeChild(btn);
< <button>NEW BUTTON!!!</button>
> |
```

Default levels ▾ | No Issues

```
> btn
< <button>NEW BUTTON!!!</button>
> p
< > <p>...</p>
> p.insertAdjacentElement('beforebegin', btn);
< <button>NEW BUTTON!!!</button>
> p.insertAdjacentElement('afterbegin', btn);
< <button>NEW BUTTON!!!</button>
> p.insertAdjacentElement('beforeend', btn);
< <button>NEW BUTTON!!!</button>
> p.insertAdjacentElement('afterend', btn);
< <button>NEW BUTTON!!!</button>
> let body = document.querySelector('body');
< undefined
> body.removeChild(btn);
< <button>NEW BUTTON!!!</button>
> p.insertAdjacentElement('afterend', btn);
< <button>NEW BUTTON!!!</button>
> btn.remove();
< undefined
> p.remove();
< undefined
```